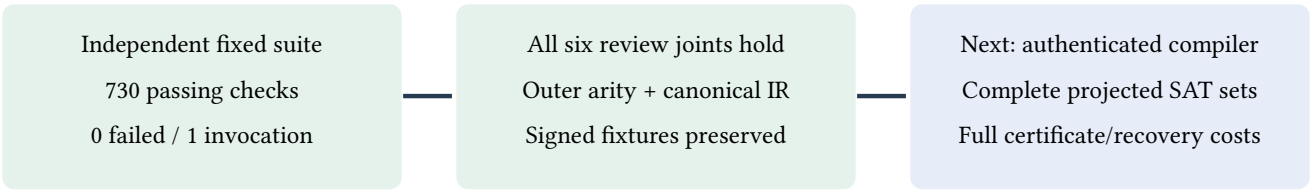
# Authenticated rationality: independent review

**DEC-20261031-ca1521 · TASK-20261031-f1574a**




TASK-20261031-c78dc8 · assertion counts, not performance units

The corrected authenticated interface passed 730 independently authored checks in one retained suite, with zero failures. Decision: support this interface at its fixed software scope. All six assigned review joints hold: outer arity, canonical authentication, symbolic binding, signed fidelity, fixed execution and claim boundary.

The source requires an exact integer k in 0..5 and equality with degree(U) before stratum selection. The R1 degree-four-to-three mutation rejects before any canonical builder call. Forty-six outer-arity checks, 128 canonical-authentication checks and 207 symbolic-binding checks passed. Eight additional controls show that the generic evaluator accepts altered or fixture-specialized equations while canonical authentication rejects them. Assignment parsing and arithmetic residual failures remain distinct from content-authentication failures.

All 15 base chart fixtures were retained: 13 valid and two impossible reference cases, with 39 valid and six invalid candidate applications across three arms. Both F1 guard roots recover exactly 4(0,1)+(0,4); swapped or balanced substitutions reject. Repeated and branch roots, characteristic-sized multiplicity, wrapped intervals, zero-B charts and target outside the interval remain covered. A group-sum test alone would not detect the swapped F1 divisor; exact signed multiplicity checks do.

The reviewer supplied independent polynomial/CRT witness arithmetic, sparse symbolic expansions, coefficient residual evaluation and signed expectations. All 14 dispatched input hashes and the executed test hash remained unchanged. Root matched the four sealed output hashes and recomputed the 730-row summary from retained raw stdout without rerunning the suite. Native review-plan/attestation consistency passed. That gate does not authenticate the serving model or prove mathematical truth; the approved inference amendment retains the runtime uncertainty.

The checked nilpotent variants k=0..5 retain 0, 3, 9, 15, 21, 36 auxiliaries and equations; guard has 10 auxiliaries and 12 equations, slack 9 and 8. These include modular quotient variables and describe symbolic field circuits, not solving work. Authentication construction/comparison events, residual arithmetic and reference checks remain separate units in the raw record.

Two native checkpoints completed successfully. The first was retained after a pre-execution test edit; the second covered the unchanged final source before the single 240-second/2GiB-bounded suite. The suite returned zero with empty stderr. No observation timeout caused a duplicate launch, and no scientific rerun was performed. The raw source, command, outputs, all rows and both receipts are retained.

This resolves the two interface obligations from DEC-20261031-cd6460 for the reviewed additive wrapper. Trust in the public descriptor and primality remains explicit; canonical-builder fidelity is a separate mathematical dependency checked here at the stated F5/F7 fixed scope. The next work is authenticated bit compilation, canonical-range certificate completion and the full projected SAT comparison. Preserve all chart strata, finite-target sign and reconstruction/certificate costs. The prospective direct-root simplifications remain separate unreviewed constructions, not part of this acceptance.

Evidence publication: b43609c3f1aa16d79e651144f2c0c89a82917a78. Exact sources: report-inputs.json.